

# JOANNE HOAR, M.Sc.

Software Developer • Technology Educator • AI Practitioner

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xcodegirl

Hanna, Alberta, Canada • Remote

## Summary

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Canadian Software Developer and Technology Educator with a Master of Science in Computer Science (University of Calgary) and more than twenty years of experience building technical systems and teaching. Currently an active online instructor at Robertson College, where I develop curriculum and teach across twelve programming and data subjects on Brightspace/D2L. I use AI tools daily across content development, code generation, CLI workflows, and process automation, and I teach students how to use AI responsibly, as a tool rather than a solution. I have developed and actively deliver curriculum on AI ethics, responsible use, and academic integrity as it applies to AI-generated content.

## AI Expertise

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### Daily practice:

- Generative AI for content development: drafting course materials, case studies, scaffolded exercises, and code examples, always with accuracy review and editorial refinement
- Prompt engineering for structured output: system prompts, iterative refinement, chain-of-thought elicitation, output format and style control
- AI-assisted coding: **GitHub Copilot**, **Claude**, **ChatGPT**: for scaffolding, debugging, refactoring, and code review
- CLI and workflow automation: using AI to interface with **Azure**, **GitHub**, and shell tooling
- Multi-model practice: Claude, ChatGPT, GitHub Copilot, DeepSeek, Recraft AI, with different tools for different tasks

### What I teach about AI:

- Responsible AI use as professional practice: practitioners are ultimately accountable for everything they produce, with or without AI assistance
- **Three-level AI assessment framework** (adapted from Perkins et al., 2024): from no-AI to AI-assisted task completion with mandatory human evaluation, applied directly to programming assignments
- Citation of AI-generated code: inline comment standards, README disclosure, terms of service compliance, with real code examples across seven scenarios
- Copyright, bias, hallucination detection, and institutional policy: applied to software contexts, not just essay writing
- AI tools as a first draft requiring editorial judgment, not a finished product

**AI and assessment design** (from *Robertson's academic integrity training and the Academic Integrity Inter-Institutional Meeting, AIIM 2026*):

- "Authentic" learning and assessment: assignments grounded in real, local contexts, where generic AI output cannot succeed without genuine human judgment
- Scaffolded assignments with checkpoints: milestone grading and visible work in progress rather than a single final deliverable
- Developing professional judgment of AI output: students audit, correct, and debug flawed, biased, or hallucinated AI drafts
- Peer review and peer grading of AI-assisted work against concrete rubrics

## Teaching & Course Development

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### Technology Instructor

Jan 2025 – Present

Robertson College – Online, Brightspace / D2L (Full Stack Development & Data Analytics programs)

- Design, develop, and maintain learning materials for **20 course offerings across 12 subjects**: weekly instructional content, sample code repositories, reference sheets, recorded walkthroughs, and open-source GitHub archives
- Teach students to use AI responsibly in software development; deliver academic integrity curriculum on AI citation, policy, and ethics; developed a dedicated slide deck actively used each term
- Completed Robertson's academic integrity training; participated in the Academic Integrity Inter-Institutional Meeting (AIIIM 2026)
- Conduct 3+ synchronous live coding sessions per week; use real-time debugging as deliberate instructional modelling of professional problem-solving
- Apply Universal Design for Learning throughout: captions on recordings, structured documents, hardware-agnostic assignments
- Student satisfaction: **4.5 / 5.0** teaching career-changing adult learners; feedback highlights real-world context, approachability, and responsiveness

**Brightspace/D2L • Angular • React • ASP.NET Core • C# • Java • Git • Azure • SQL • Python**

### Instructor, Intermediate C++ Programming

2005 – 2008

University of Calgary, Department of Continuing Education – Adult learners

- Designed and delivered complete curriculum for adult learners: outcomes, lectures, sample code, and assessments from the ground up

### Research / Teaching Assistant

2001 – 2004

VLab / AI Research Hub, University of Calgary

- TA and lab instructor: *Introduction to Programming, Computing for Non-Majors*, and *Operating Systems*
- Computing for Non-Majors required translating technical concepts for a non-technical audience, directly transferable to AI for Business Leaders contexts

## Industry Experience

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### Senior Software Developer

2022 – 2025

bitHeads Inc. – brainCloud BaaS SDK

- Authored extensive developer-facing documentation, onboarding guides, and sample projects; hired in part for educational background
- Provided one-on-one client instruction across seven UI frameworks and twelve platforms

### Co-founder / Lead Developer

2005 – 2022

Silicon Hanna Inc. (self-employed; founded as SuRJE Software Solutions Inc.)

- 17-year software development and consulting practice spanning mobile, web, AR, and games
- Early AI/ML research roots: M.Sc. in agent-based systems, swarm intelligence, and evolutionary computation, the academic precursors to today's machine learning landscape
- Published 11 mobile titles (iOS and Android); explored novel SDKs and produced learning materials throughout

### Tools Programmer

2008 – 2011

Fekete Associates Inc. (S&P Global)

- Designed and delivered a voluntary **internal graphics training course** for colleagues from beginners to senior developers; built multi-level materials with a hands-on capstone

### Intermediate Software Engineer

2005 – 2008

*BJ Pipeline Inspection Services Inc. (Baker Hughes)*

- Implemented **neural network algorithms** for automated anomaly detection in pipeline inspection data: applied machine learning in industry well before the current AI wave

## Education

- **M.Sc. Computer Science** – University of Calgary 2004  
*Thesis: Agent-based Development of Natural Transportation Networks*  
*Graduate coursework: Human-Computer Interaction, Artificial Intelligence, Computer Graphics*
- **B.Sc. Computer Science** – University of Calgary 2001

## Honours & Awards

- Departmental Nomination: Governor General's Gold Medal, Faculty of Graduate Studies 2005
- Postgraduate Scholarship NSERC D3, Natural Sciences and Engineering Research Council of Canada 2005
- Graduate Student Scholarship, Alberta Learning 2004
- Best Student Paper Award (Co-author), Australian Conference on Artificial Life (ACAL) 2003
- CPSC Departmental Research Award, University of Calgary 2003

## Selected Publications *(published under maiden name Penner)*

- **Penner, J.**, Hoar, R., Jacob, C. "Bacterial Chemotaxis in Silico," *Australian Conference on Artificial Life (ACAL)*, 2003  
*Best Student Paper*
- Hoar, R., **Penner, J.**, Jacob, C. "Transcription and Evolution of a Virtual Bacteria Culture," *IEEE Congress on Evolutionary Computation (CEC)*, 2003
- Hoar, R., **Penner, J.** "The Application of Artificial Intelligence to Transportation System Design," *XRDS: Crossroads, ACM*, 2003
- **Penner, J.**, Hoar, R., Jacob, C. "Swarm-Based Traffic Simulation with Evolutionary Traffic Light Adaptation," *IASTED ASM*, 2002
- Hoar, R., **Penner, J.**, Jacob, C. "Evolutionary Swarm Traffic: If Ant Roads Had Traffic Lights," *IEEE CEC*, 2002

## Technical Skills

**AI Tools:** ChatGPT, Claude, GitHub Copilot, DeepSeek, Recraft AI, used responsibly in curriculum, content, and coding workflows

**LMS:** Brightspace / D2L (daily), GitHub (open-source course repos)

**Languages:** C++, C#, JavaScript, TypeScript, Java, Python, PHP, SQL

**Frameworks:** Angular, React, ASP.NET Core, Node.js, .NET

**Content & Authoring:** Markdown,  $\text{\LaTeX}$ , HTML/CSS, open-source tooling

**Human Languages:** English (native), French (intermediate), Spanish (rudimentary)

## Work Samples

 [github.com/xcodegirl/career](https://github.com/xcodegirl/career) (xcodegirl-work-samples.md)

Academic integrity / responsible AI use slide deck available on request.